
Is Tylenol the same as paracetamol, is it safe during pregnancy and does it have links to autism?

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Is Tylenol the same as paracetamol, is it safe during pregnancy and does it have links to autism?

Donald Trump has [urged pregnant women not to take acetaminophen](#), also known as Tylenol or paracetamol. He claimed it raises the chances of children being autistic.

But the US president has been [condemned by experts from across the world](#), who fear he is deliberately fostering a narrative of distrust that could be dangerous for women.

So what does the science *actually* say?

Does paracetamol cause autism?

The short answer is: no. Genetics plays a major role in autism, but scientists are also investigating the potential role of environmental factors such as parental age, premature birth and prenatal exposure to drugs, environmental chemicals or infections.

"The idea that something that occurs during pregnancy may influence neurodevelopment isn't a particularly new or crazy or outlandish idea," said Renee Gardner, an associate professor of epidemiology at Sweden's Karolinska Institute. "We know there's a large heritable or familial influence, but environment interacts with genetic tendencies."

Paracetamol has been widely used for pain relief since the 1950s and is recommended by the NHS as the first choice of painkiller if you are pregnant. In recent years, [a number of studies](#) have suggested that women who take paracetamol during pregnancy are very slightly more likely to have a child who is diagnosed with autism. But this does not prove that the link is causal and agencies such as the UK's Medicines and Healthcare products Regulatory Agency (MHRA) have stated there is no evidence that taking paracetamol during pregnancy causes autism in children.

Why do some studies suggest a link between paracetamol and autism?

A big challenge in human studies is working out what genuinely contributes to autism and what is simply associated with the condition.

"Lots of potential risk factors for autism have been investigated and when you are doing this kind of research, it can be very difficult to disentangle what is truly a cause," said Prof Laurie Tomlinson, professor of clinical epidemiology at the London School of Hygiene and Tropical Medicine. "People taking paracetamol or not taking paracetamol in pregnancy may be different in many ways."

For example, older women are more likely to have autistic children, but they are also more likely to have aches and pains and medical conditions that call for painkillers.

Similarly, particular medical conditions are biologically connected with autism. For example, women with hypermobility have loose and stretchy ligaments that mean their joints have an unusually large range of motion. They are also more likely to have autistic children. Pregnancy can cause worse joint pain in women with hypermobility, making them more likely to take painkillers. If studies do not

fully account for these factors, painkillers can wrongly appear to be a risk factor.

Gardner and colleagues led [the largest study to date on autism and paracetamol use during pregnancy](#), using health records of 2.4 million children born in Sweden between 1995 and 2019. They found that about 1.5% of children whose mothers were prescribed paracetamol during pregnancy were autistic, compared with 1.3% of children whose mothers were not prescribed the drug.

However, when the researchers compared outcomes between siblings – removing genetics from the equation – the association between autism and paracetamol use disappeared completely. "Some of the same genetic tendencies that increase people's risk of autism are overlapping with genes that influence immune response and pain perception, which are all things that might increase use of paracetamol," said Gardner.

The researchers concluded there was no evidence to support a causal link and another Japanese study, published earlier this year, reached a similar verdict.

Why are autism rates rising?

Forty years ago, about four to six people in every 10,000 had an autism diagnosis. Today, depending on the country, the rate is somewhere between 1% and 3%. Diagnosis has increased in adults as well as young children. Trump's health secretary, Robert F Kennedy Jr, has described the increase as an "epidemic", a word better reserved for infectious disease outbreaks.

Many researchers believe the rise is because of doctors casting a wider net when diagnosing autism, greater awareness, and clinics improving diagnosis, rather than an increase in autism itself. But it is possible that as-yet undiscovered environmental factors have also contributed.

"A much broader range of behaviours were brought under the umbrella of the term autism," said Prof Gina Rippon, emeritus professor of cognitive neuroimaging at Aston University. "What looked like a dramatic increase in cases was a dramatic increase in diagnosis."

"The whole impression is that this is a disease that needs curing or eradicating," she added. "It's putting aside the huge advantages of the creativity and inventiveness of people who think about the world in a different way. To paint it as an epidemic or disease is very demeaning to the autistic community."

Trump's comments 'feed a narrative of distrust ... which is really dangerous for women'

The best evidence does not support the claim that paracetamol contributes to autism, but Trump's comments will still have a wide impact. One immediate effect will be more pain and misery for women in pregnancy, with potential health risks.

"When Trump suggests that there's no downside to not taking paracetamol during pregnancy, that's not true," said Gardner. "There's quite a big body of evidence that shows that there are potentially severe health risks if a fever goes untreated for a prolonged period of time during pregnancy."

There is also a broader undermining of trust in science and the medical establishment. "Pregnant women are already bombarded with a host of unsubstantiated information," said Prof Sarah Hawkes, the co-founder of Global Health 50/50, a gender equality research initiative. "It feeds into a narrative of distrust in science, which is really dangerous for women's health, particularly in pregnancy."

For parents of autistic children, Hawkes added, "you've introduced a huge amount of stigma and guilt when the scientific evidence just doesn't bear that out".

There is a risk to women's mental health, too. Prof Shakila Thangaratinam, professor of women's health at the University of Liverpool, said the advice was likely to make women more anxious while pregnant and after childbirth. "Women could spend the next 10 years unnecessarily worrying," she said.

Given the power and influence that US health institutions have had as standard-setting agencies, advice issued by the FDA is likely to have consequences around the world. In some countries, paracetamol is the primary painkiller given in childbirth. If doctors or pregnant women avoid the painkiller, it could lead to more post-traumatic stress disorder and to more women seeking caesarean sections.

"Paracetamol is far and away the safest painkiller in pregnancy so if women are more reluctant to take it they are going to be suffering unnecessarily," said Tomlinson. "In general, women are very cautious about taking anything during pregnancy."

What will change in the US?

The [US Food and Drug Administration is changing the label](#) on Tylenol to warn of a "possible association" between autism and taking the drug in pregnancy. It has also alerted physicians nationally. The agency makes clear that a causal relationship between Tylenol and autism has not been established. It also notes that it is the only drug approved to treat fever in pregnancy and high fever can be risky for babies in the womb.

But experts are concerned that Trump's less measured instruction to "fight like hell not to take it" may be what ends up sticking in the minds of many patients.

"Can you imagine having to sit there and weigh the decision?" said Gardner. "Your doctor may be telling you it's safe to use, but the bold words on the packaging say this could cause harm to your unborn baby. That's a terrible position to put parents into."

The FDA also moved to approve a drug called leucovorin, a form of folate, amid research that suggests it may improve some of the social communication difficulties autistic people can experience.

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